Generative AI HW4 電機四 4111064113 白舜文

Ⅰ. Problem Description

1. The attached python program can generate 16 face images from a pre-trained StyleGAN model. Since we know that fixing a portion of latent variables will limit the variations of generated faces, please try to make the first 128 elements of the 512-d latent variables all the same and generate 16 faces. After this experiment, make the first 384 elements of the latent variables all the same and generate another 16 faces. Hand in your codes and the two sets of faces you generated. (In fact, when we begin to limit the randomness of latent variables in StyleGAN, artifacts in generated faces will become more prominent. You can observe this phenomenon in this experiment.)

II. StyleGAN results

fixed_128.png:

fixed_384.png:

根據上面兩種實驗可知，透過潛在變數的改變，並限制其隨機性，可對於影像生成的品質進行初步的探討：

1. 在固定前 128 維潛在變數下，人臉依然保持著高度的多樣性，包含配件、膚色、髮型等等特徵，整體的相似度不高，圖像也較為自然。

2. 接著限制到 384 維潛在變數，可觀察到整體的相似性提高許多，包含臉部角度、臉型特徵、衣服顏色等等。此時僅剩 128 維在控制細節導致變異性降低，同時也可觀察到圖像中的 Artifacts(圖形的偽影、瑕疵)也變多了，在影像邊緣的雜訊斑點變多，頭髮的銜接處也明顯的不自然。

III. Source code

```
import torch
from matplotlib import pyplot
import torchvision

model = torch.hub.load('ndahlquist/pytorch-hub-stylegan:0.0.1', 'style_gan', pretrained=True)

# %matplotlib inline

model.eval()
device = 'cuda:0' if torch.cuda.is_available() else 'cpu'
model.to(device)


def stylegan_experiment(fix_dim, save_name):
    nb_rows = 4
    nb_cols = 4
    nb_samples = nb_rows * nb_cols

    latents = torch.randn(nb_samples, 512, device=device)

    # replace the fix_dim elements in 1~15 samples with element corresponding to the 0 sample
    if fix_dim > 0:
        latents[1:, :fix_dim] = latents[0, :fix_dim].clone()

    with torch.no_grad():
        imgs = model(latents)
        # normalization to 0..1 range
        imgs = (imgs.clamp(-1, 1) + 1) / 2.0

    imgs = imgs.cpu()
    grid_img = torchvision.utils.make_grid(imgs, nrow=nb_cols)

    pyplot.figure(figsize=(10, 10))
    pyplot.imshow(grid_img.permute(1, 2, 0).detach().numpy())
    pyplot.title(f"StyleGAN: {fix_dim} elements fixed")
    pyplot.axis('off')
    pyplot.savefig(save_name)

# Experiment 1: first 128 elements remain same
print("Experiment 1 executed: fixed_128 elements")
stylegan_experiment(128, "fixed_128.png")

# Experiment 2: first 384 elements remain same
print("Experiment 2 executed: fixed 384 elements")
stylegan_experiment(384, "fixed_384.png")

pyplot.show()
```